

Red MMO — Developer Handoff (EN)

2026-08-07. Same-repo collaboration: work in HubLogic (L_Hub_Gameplay_Logic); artists own L_Hub_Env_Visuals; thin L_Hub_Persistent composes both. Clone RedMMOTitan-DeveloperHandoff → TitanFundamentals. Netcode: NETWORKING_PVP.md.

START HERE — Red MMO developer handoff

Generated: 2026-08-07
Ukrainian: START_HERE_UK.md
PDF (EN): RedMMOTitan_Developer_Handoff.pdf · PDF (UK): RedMMOTitan_Developer_Handoff_UK.pdf
Product name in docs: Red MMO. Names like TitanFundamentals.uproject / RedMMOTitan* are technical paths only.

Core: same repo, merge-safe world (read this first)

You and the environment artists work in the same GitHub repository. You will not lose your hub gameplay when the player-hub environment is built, and you will not have to reprogram everything after art lands — if you keep ownership split:

Layer	Package	Owner
Thin persistent	/Game/RedMMO/Maps/Hubs/L_Hub_Persistent	Lead / rare edits
Environment	/Game/RedMMO/Maps/Hubs/L_Hub_Env_Visuals	Environment artists
HubLogic (you)	/Game/RedMMO/Maps/Hubs/L_Hub_Gameplay_Logic	Gameplay developer

- Put PlayerStarts, volumes, spawn managers, replicated / networked actors in L_Hub_Gameplay_Logic.
 - Do not edit the artist-owned env .umap.
 - Keep C++ / gameplay Blueprints in your packages — they are independent of env map saves.
 - Details: MERGE_SAFE_WORLD.md · artist merge doc: ../EnvironmentArtistHandoff/MERGE_ENV_AND_GAMEPLAY.md.
- One-liner: Clone this handoff repo, work in the HubLogic / gameplay layer; artists will not overwrite your code or gameplay map.

60-second path

- 1 Clone: <https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git>
 - 2 Install Unreal Engine 5.8.
 - 3 Open TitanFundamentals.uproject (PlanetGen / PPG not required).
 - 4 Build TitanEditor Win64 Development.
 - 5 Day-1 PIE: /Game/ThirdPerson/Maps/ThirdPersonMap (controls / listen-server).
- Hub work that must survive env delivery: L_Hub_Gameplay_Logic under /Game/RedMMO/Maps/Hubs/.
- 1 Netcode track: NETWORKING_PVP.md. Full pack: DEVELOPER_HANDOFF.md.
- ```
git clone https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git
cd RedMMOTitan-DeveloperHandoff
start TitanFundamentals.uproject
```

| Check                               | Expected                                                  |
|-------------------------------------|-----------------------------------------------------------|
| Editor opens without planet plugins | No hard fail for missing PPG / PlanetGen                  |
| PIE on ThirdPersonMap               | Character moves; listen-server 2P for replication tests   |
| Hub stubs present                   | L_Hub_Persistent, L_Hub_Env_Visuals, L_Hub_Gameplay_Logic |
| Env sublevel                        | You do not save artist ownership into that package        |

|              |                                                       |
|--------------|-------------------------------------------------------|
| Running anim | Do not overwrite trooper locomotion without authority |
|--------------|-------------------------------------------------------|

## What to read next

| Doc                          | Why                                           |
|------------------------------|-----------------------------------------------|
| MERGE_SAFE_WORLD.md          | Same-repo ownership + no-overwrite merge path |
| NETWORKING_PVP.md            | Listen-server, replication, PvP order of work |
| DEVELOPER_HANDOFF.md         | Paths, protected items, plugin disposition    |
| PPG_PLANETGEN_FREE_START.md  | Fundamentals without planet plugins           |
| FAB_MARKETPLACE_INVENTORY.md | Optional marketplace plugins later            |

Environment artist role (separate): ../EnvironmentArtistHandoff/START\_HERE.md

## Protected (short list)

- Running / locomotion on ABP\_RedTrooperFemale
- Artist-owned env map packages (do not treat as your hub file)
- Do not enable Steam Integration Kit and Epic OnlineSubsystemSteam together

# Same-repo, merge-safe world ownership

This is the core collaboration rule for Red MMO.

Gameplay and environment work in the same GitHub repo, on separate Unreal map packages, so hub art can land without forcing a gameplay rewrite and without binary .umap overwrite fights.

Full artist-side detail: ../EnvironmentArtistHandoff/MERGE\_ENV\_AND\_GAMEPLAY.md.

## What you are guaranteed

| Need                                                                        | How it is met                                                                                                                                           |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| A world/map setup that is not lost when the player hub environment is built | Your gameplay actors live in L_Hub_Gameplay_Logic (HubLogic). Env dress lives in L_Hub_Env_Visuals.                                                     |
| You do not reprogram everything after env artists finish the hub            | C++ / Blueprint assets (Character, GameMode, weapons, AnimBP) are separate packages from env .umap files. Art swaps the env sublevel; your logic stays. |
| Same-repo collaboration that does not overwrite the other discipline        | Artists never check in the gameplay sublevel; you never edit the env sublevel. Thin persistent level only references both.                              |

## Map layout (fundamentals starter)

/Game/RedMMO/Maps/Hubs/L\_Hub\_Persistent  
/Game/RedMMO/Maps/Hubs/L\_Hub\_Env\_Visuals  
/Game/RedMMO/Maps/Hubs/L\_Hub\_Gameplay\_Logic

← thin shell (streaming refs only; rarely edited)  
← environment artists  
← you (HubLogic): PlayerStarts, volumes, replicated actors

| Owner                        | Owns                                                        | Do not touch                                    |
|------------------------------|-------------------------------------------------------------|-------------------------------------------------|
| Gameplay developer           | L_Hub_Gameplay_Logic, Source/, gameplay Blueprints, netcode | Artist-owned env .umap                          |
| Environment artist           | L_Hub_Env_Visuals, landscape/lighting/foilage/dressing      | Gameplay sublevel, Character/GameMode packages  |
| Lead / either with agreement | L_Hub_Persistent membership (which sublevels load)          | Daily art or combat authoring inside Persistent |

Day-1 compile / listen-server PIE can still use /Game/ThirdPerson/Maps/ThirdPersonMap. Put hub gameplay that must survive env delivery into L\_Hub\_Gameplay\_Logic.

## Working rules

- 1 Clone the handoff repo; open TitanFundamentals.uproject.
- 2 Put hub PlayerStarts, combat volumes, spawn managers, networked actors in L\_Hub\_Gameplay\_Logic.
- 3 Do not dress the long-term hub look inside the gameplay sublevel or inside a single shared HubWorld.umap.
- 4 Do not edit L\_Hub\_Env\_Visuals (or ArtistCanvas / sandbox env delivery maps) as your ownership file.
- 5 When env art is ready, Persistent loads both sublevels — you fix only cross-level references (streaming, soft paths), not a full rewrite.

Binary truth: two people saving the same .umap cannot be safely git-merged. Split packages first.

## One sentence for the developer

Clone the handoff repo, work in the HubLogic / gameplay layer (L\_Hub\_Gameplay\_Logic + C++/BP); artists own the env sublevel and will not overwrite your code or gameplay map.

# Networking / Replication / Multiplayer / PvP — programmer start

Audience: gameplay / netcode programmer (listen-server, replication, PvP).

Ukrainian: NETWORKING\_PVP\_UK.md

PPG/PlanetGen: not required for first netcode PIE — use `TitanFundamentals.uproject` + `ThirdPersonMap`.

## World ownership (short)

Same GitHub repo as environment artists. Put hub gameplay / networked actors in

`/Game/RedMMO/Maps/Hubs/L_Hub_Gameplay_Logic` (HubLogic). Do not edit `L_Hub_Env_Visuals`. Thin `L_Hub_Persistent` composes both so env delivery does not force a netcode rewrite. Full rules: `MERGE_SAFE_WORLD.md`.

## Current architecture (honest)

| Layer      | What exists                                                                                                                                                                                                                          | Proven?                                                                                                                 |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Topology   | Unreal listen-server (host is also a player). Dedicated server not accepted.                                                                                                                                                         | PIE 2-client listen-server used historically for aim/jetpack/fire replication                                           |
| Transport  | Steam Integration Kit (SIK) v1.9 as sole Steam stack (replacement <code>OnlineSubsystemSteam</code> + <code>SteamSockets</code> ). Epic stock OSS Steam plugins disabled. Dev App ID 480 (Spacewar) via <code>steam_appid.txt</code> | Packaged smokes: Steam auth/relay + single-account host lifecycle. Real two-account Steam join/PvP not proven           |
| Session UI | Escape → Multiplayer/Lobby; F8 runtime / F6 in PIE (editor owns F8). Create / Find / Join / Reconnect / Invite / Leave. Sessions advertise ≥8 players. No silent auto-host after empty search                                        | UI + host lifecycle smoke; friend join open                                                                             |
| Authority  | Server-authoritative weapons heat/overheat, bolt spawn, ship pilot/weapons, damage; aim direction + jetpack flags replicated                                                                                                         | Two-client listen-server PIE verified aim/jetpack/fire/heat (NULL OSS path — gameplay replication, not Steam transport) |
| GameState  | Celestial / PPG frame producers and registry work exist; real 1S/2C late-join transport + radar consumers open (M05)                                                                                                                 | Single-process ordering evidence ≠ multiplayer acceptance                                                               |
| PvP        | <code>bIsEnemy</code> , health/shield/armor, downed, grapple-on-player pull, lethal ragdoll / ship death paths in character/ship code                                                                                                | Not a finished PvP mode; no ranked/matchmaking                                                                          |

## Key classes (start here)

| System                           | Primary files                                                                                                | Net notes                                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Player / jetpack / aim / weapons | <code>Source/RedMMO/RedPlayerCharacter.h/.cpp</code>                                                         | Large <code>DOREPLIFETIME</code> block: Health/Shield/Armor/Fuel, aim, weapon heat, jetpack flags, abilities, grapple/slam |
| Movement                         | <code>RedCharacterMovement.*</code>                                                                          | Radial gravity / planet movement — test carefully under listen-server                                                      |
| Projectiles                      | <code>RedBolt.*</code> , <code>WeaponFirer.*</code>                                                          | Server-spawned bolts; presentation vs authority                                                                            |
| Ship (StarSparrow)               | <code>RedShip.*</code> , <code>RedShipMovementComponent.*</code>                                             | Replicated Health, WeaponHeat, Pilot, landing flags                                                                        |
| Shuttle                          | <code>RedShuttleBase.*</code>                                                                                | Related craft occupancy / audio GC fix history                                                                             |
| Sessions / Steam                 | <code>RedGameInstance.*</code> , <code>RedSessionBrowserWidget.*</code> , <code>RedHUD.*</code> (lobby open) | SIK session browser; App 480                                                                                               |
| Pause / lobby entry              | <code>RedPauseMenuWidget.*</code>                                                                            | Escape menu Multiplayer row                                                                                                |

|                          |                                                         |                                                                       |
|--------------------------|---------------------------------------------------------|-----------------------------------------------------------------------|
| GameMode                 | RedGameMode.*, clean project RedPPGGameplayGameMode.cpp | Default modes differ Titan vs clean RedMMO                            |
| Celestial / MP consumers | RedCelestialFrameRegistry.*, GameState helpers          | M05 still open for real 1S/2C                                         |
| Focused MP tests         | Source/RedMMO/Tests/RedDEF0003TwoClientPIETests.cpp     | Listen-server count helpers — extend, don't treat as full Steam proof |

Anim / aim presentation (protect run): Content/RedMMO/Characters/ABP\_RedTrooperFemale — do not casually rewrite run/locomotion. Aim/jetpack pose work should stay additive / parallel. FocalRig marketplace absent → ControlRig substitute in RedMMOEditorTools.

## Where to start (recommended order)

- Day 1 — no Fab planets: Open TitanFundamentals.uproject, build TitanEditor, PIE ThirdPersonMap. Confirm compile with PlanetGen/PPG/FocalRig/WorldGen off.
- Day 1–2 — listen-server gameplay replication: Editor → Play → Number of Players = 2, Net Mode = Play As Listen Server. Validate move/aim/fire/jetpack replication between windows (NULL OSS is fine for this gate).
- Day 2–3 — PvP damage loop: Trace Health/Shield damage, bIsEnemy, downed, bolt hit authority in RedPlayerCharacter / RedBolt. Add a minimal deathmatch spawn/score only after replication is stable.
- Hub placement: keep networked hub actors / PlayerStarts that must survive env delivery in L\_Hub\_Gameplay\_Logic, not in the env sublevel.
- When ready for Steam: Enable/copy Plugins/SteamIntegrationKit, keep Epic OSS Steam disabled, Steam client logged in, App 480. Host on machine A, join on machine B with different Steam accounts. Do not claim success from single-account smoke.
- Defer: PPG home world, PlanetGen maps, fused terrain, final art — not blockers for netcode fundamentals.

## Known gaps (net / MP)

- Real 1 server + 2 clients Steam transport (initial/update/stale/late-join) unaccepted
- Dedicated server, persistence, matchmaking, Steam App ID beyond 480
- Radar/minimap multiplayer consumers
- Warp / interplanetary replication
- Packaged two-account PvP
- Clean RedMMO vs Titan session parity

## Protected items

- Running animation / locomotion single-node guard on trooper ABP
- R92 package tree identity; fused 27/6 authoring hashes
- Do not enable both SIK and Epic OnlineSubsystemSteam together

## Fab / plugins for this track

See FAB\_MARKETPLACE\_INVENTORY.md.

| Need                     | Plugin                                                                           |
|--------------------------|----------------------------------------------------------------------------------|
| Netcode PIE fundamentals | None from Fab                                                                    |
| Steam friend sessions    | SteamIntegrationKit (project plugin; ~297 MB — may need copy from owner machine) |

|                |                                  |
|----------------|----------------------------------|
| Planetary maps | PPG / PlanetGen — optional later |
|----------------|----------------------------------|

# Red MMO — Developer Handoff (2026-08-07)

Status: Good enough to pick up. Many problems remain. Fundamentals do not require planet plugins.

Ukrainian: DEVELOPER\_HANDOFF\_UK.md · START: START\_HERE.md

Audience: gameplay / netcode programmer. World ownership first: MERGE\_SAFE\_WORLD.md. Netcode order: NETWORKING\_PVP.md.

| Artifact         | Location                                                                                                                                                    |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| START_HERE       | START_HERE.md · START_HERE_UK.md                                                                                                                            |
| Merge-safe world | MERGE_SAFE_WORLD.md · MERGE_SAFE_WORLD_UK.md                                                                                                                |
| PDF EN / UK      | RedMMOTitan_Developer_Handoff.pdf · RedMMOTitan_Developer_Handoff_UK.pdf                                                                                    |
| Networking / PvP | NETWORKING_PVP.md · NETWORKING_PVP_UK.md                                                                                                                    |
| PPG-free start   | PPG_PLANETGEN_FREE_START.md                                                                                                                                 |
| Fab inventory    | FAB_MARKETPLACE_INVENTORY.md                                                                                                                                |
| Clone            | <a href="https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git">https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git</a> |

## 1. Same-repo collaboration (non-negotiable)

Gameplay and environment share one repository. Separable ownership keeps hub art from forcing a gameplay rewrite:

```
L_Hub_Persistent ← thin shell (streaming only)
■■■ L_Hub_Env_Visuals ← artists (do not edit)
■■■ L_Hub_Gameplay_Logic ← you / HubLogic
```

| You own                                                   | Artists own                                                   |
|-----------------------------------------------------------|---------------------------------------------------------------|
| L_Hub_Gameplay_Logic, Source/RedMMO, gameplay BP, netcode | L_Hub_Env_Visuals, landscape, lighting, foliage, set dressing |

Guarantees: hub gameplay map package is not the env delivery file; C++/BP are not rewritten when env .umap changes; git will not “merge” two writers on one binary .umap — so do not share one.

Aligned with ../EnvironmentArtistHandoff/MERGE\_ENV\_AND\_GAMEPLAY.md.

## 2. Paths

| Role                            | Path                                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Standalone handoff (clone this) | <a href="https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git">https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git</a><br>→ TitanFundamentals.uproject |
| Hub map stubs                   | /Game/RedMMO/Maps/Hubs/ (L_Hub_Persistent, L_Hub_Env_Visuals, L_Hub_Gameplay_Logic)                                                                                                         |
| Day-1 PIE (no hub art required) | /Game/ThirdPerson/Maps/ThirdPersonMap                                                                                                                                                       |
| Engine                          | Unreal Engine 5.8                                                                                                                                                                           |

Optional later (owner / full content): full Titan.uproject planetary maps, clean D-drive RedMMO.uproject + PPG home world, packaged R92 baseline — not required for HubLogic / netcode start.

## 3. What works / what is open (honest)

Working enough to start:

- TitanEditor Win64 Development with FocalRig/WorldGen disabled and PlanetGen APIs shimmed (RedPlanetGenCompat)

- Fundamentals PIE without PPG / PlanetGen
- Listen-server aim / fire / jetpack replication (gameplay path; not a finished Steam two-account proof)
- Live RedMMO editor fixes historically: wheel→thrust, atmosphere exit, weapons/plumes/jetpack, terrain blend (re-verify on your build)

Still open:

- Physical feel polish; HUD fuel pixels; biome art; fused consumer; M04/M05; Steam two-account PvP; dedicated server
- Composed hub Persistent with real env + gameplay streaming (stubs exist; dress and wire on integration days)

## 4. Build / PIE

```
git clone https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git
cd RedMMOTitan-DeveloperHandoff
start TitanFundamentals.uproject
Target: TitanEditor | Win64 | Development
```

- 1 Confirm compile with PlanetGen / PPG / FocalRig / WorldGen off.
- 2 PIE ThirdPersonMap for controls and 2-player listen-server.
- 3 Author hub gameplay in L\_Hub\_Gameplay\_Logic; leave L\_Hub\_Env\_Visuals to artists.
- 4 Wire both under L\_Hub\_Persistent when integrating.

One UnrealEditor at a time on a shared machine.

## 5. Plugin disposition (fundamentals)

| Plugin    | Fundamentals                                             |
|-----------|----------------------------------------------------------|
| FocalRig  | Disabled (+ ControlRig substitute in source)             |
| WorldGen  | Disabled                                                 |
| PlanetGen | Disabled (RedPlanetGenCompat shims link)                 |
| PPG       | Not in TitanFundamentals; optional later on clean RedMMO |

See PPG\_PLANETGEN\_FREE\_START.md.

## 6. Protected items

- 1 Running animation / locomotion on ABP\_RedTrooperFemale — do not replace without explicit authority.
- 2 Env-owned map packages — not your save target for hub logic.
- 3 Fused 27/6 authoring hashes and production planetary maps — read-only without a dedicated gate.
- 4 Do not enable SIK and Epic OnlineSubsystemSteam together.

## 7. Next work order

- 1 Fundamentals compile + PIE (ThirdPersonMap).
- 2 Listen-server replication → PvP damage loop (NETWORKING\_PVP.md).
- 3 HubLogic actors in L\_Hub\_Gameplay\_Logic so env can land without rewrite.
- 4 Steam SIK two-account only after gameplay replication is stable.
- 5 Planetary / PPG maps only when those plugins are intentionally in scope.



## 8. Clone contents (expectations)

Expect: Source/, Config/, docs/ (Developer + Environment handoffs), TitanFundamentals.uproject, ThirdPerson starter map, Hub stubs under Content/RedMMO/Maps/Hubs/, selected RedMMO crumbs.

Do not expect: full Fab packs, Binaries/Intermediate/DDC, large SteamIntegrationKit tree unless copied separately, owner machine diagnostics / packaged R92 trees.

## PPG / PlanetGen—free start (required for developer onboarding)

Goal: The developer must be able to build and PIE fundamentals without installing PPG or PlanetGen.

### Verdict (2026-08-07)

| Path                                         | Needs PPG?            | Needs PlanetGen?                     | Use for                                                     |
|----------------------------------------------|-----------------------|--------------------------------------|-------------------------------------------------------------|
| TitanFundamentals.uproject + ThirdPersonMap  | No                    | No                                   | Immediate compile, controls, weapons, HUD, C++ fundamentals |
| Titan.uproject + RedPlanetGen                | No (PPG not in Titan) | Yes (enabled; home map is PlanetGen) | Full Titan planetary prototype                              |
| Clean RedMMO.uproject + RedMMO_PPG_HomeWorld | Yes                   | Reserved / separate                  | Accepted R92 planetary baseline                             |

Feasible in this handoff: Yes — use TitanFundamentals.uproject.

PlanetGen and PPG are optional for the fundamentals baseline. Full planet maps remain plugin-dependent and are documented below.

### How to start without PPG / PlanetGen

- 1 Open TitanFundamentals.uproject (not RedMMO.uproject, not full planetary maps).
- 2 Confirm plugins:
  - PlanetGen → Enabled: false
  - WorldGen → Enabled: false
  - FocalRig → Enabled: false (stock ControlRig substitute already in source)
  - No PPG entry in this uproject
- 1 Editor / game default map: /Game/ThirdPerson/Maps/ThirdPersonMap
- 2 Build TitanEditor Win64 Development → PIE.

### Shims already in project source

| Shim                   | Path                                              | Behavior when plugin/fork missing                                                          |
|------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------|
| PlanetGen query compat | Source/RedMMO/RedPlanetGenCompat.h/.cpp           | Forwards to fork APIs if headers exist; else chunk/radial substitutes so TitanEditor links |
| FocalRig optional      | Source/RedMMO/RedMMO.Build.cs + RedMMOEditorTools | REDMMO_WITH_MARKETPLACE_FOCALRIG=0 → stock ControlRig AimItem substitute                   |
| WorldGen               | Titan.uproject / Fundamentals                     | Disabled; no hard module dependency                                                        |

### What still needs plugins (optional later)

| Work                                                                             | Required                                                          |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Clean home world RedMMO_PPG_HomeWorld, ProfileV1 biomes, native PPG water/chunks | PPG marketplace plugin                                            |
| Titan RedPlanetGen / 50 km PlanetGen maps, ACLMPlanet streaming                  | PlanetGen (engine Marketplace 1.7 and/or project pin)             |
| Exact fork terrain stamp / MacroHeightfield APIs                                 | Plugins/PlanetGenPinned_* (absent on live tree; shims cover link) |

### Fallback maps

| Map | Plugin need | Notes |
|-----|-------------|-------|
|-----|-------------|-------|

|                                                 |                       |                                                                                    |
|-------------------------------------------------|-----------------------|------------------------------------------------------------------------------------|
| /Game/ThirdPerson/Maps/ThirdPersonMap           | None                  | Primary PPG/PlanetGen-free starter (included in handoff Content)                   |
| /Game/RedMMO/Maps/Hubs/L_Hub_*                  | None                  | Merge-safe hub stubs (Persistent / Env / Gameplay_Logic) — see MERGE_SAFE_WORLD.md |
| /Game/RedMMO/Maps/Sandbox_DesertDemoSparkle_T01 | Usually none for open | Local Titan sandbox; large; may reference dressed content                          |
| /Game/RedMMO/Maps/RedPlanetGen*                 | PlanetGen             | Do not use for first-day fundamentals                                              |
| /Game/RedMMO/Maps/RedMMO_PPG_Homeworld          | PPG                   | Clean D-drive project only                                                         |

If ThirdPerson content is missing after clone, create File → New Level → Empty Open World / Basic and set it as Editor Startup Map under Project Settings. That empty level is enough to compile and test C++ pawn/GameMode wiring.

## Restoring planetary plugins later

See FAB\_MARKETPLACE\_INVENTORY.md and

D:\RedMMOTitanWindowsData\Diagnostics\R92\_DependencyUnblock\_20260807\_000731\R92\_DEPENDENCY\_DISPOSITION.md

.

- 1 Install official Fab payloads so \*.uplugin exists (FocalRig/WorldGen currently Intermediate-only on owner machine).
- 2 Re-enable plugin flags in Titan.uproject.
- 3 Rebuild; FocalRig auto-links when descriptor returns.
- 4 For clean RedMMO planetary baseline, keep using D:\RedMMOTitanWindowsData\Projects\RedMMO\RedMMO.uproject with PPG enabled.

## Fab / Marketplace inventory checklist

Purpose: What came from Fab/Marketplace, whether it is required for fundamentals, and what happens if missing.

Sources: Titan.uproject, RedMMO.uproject, Plugins/, ProjectKnowledge, R92 dependency disposition.

Legend: Req = required for fundamentals (PPG-free) · Planet = required for planetary maps · Opt = optional · Off = disabled / substituted now

### Plugins / engine marketplace

| Name                                       | Purpose                                                  | Fundamentals                                  | If missing                                                                                                                                                                                                                                     | Fab / notes                                                                                                                                            |
|--------------------------------------------|----------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| PlanetGen                                  | Spherical planet (ACLMPlanet), Titan RedPlanetGen maps   | Off for fundamentals; Planet for Titan maps   | TitanFundamentals disables it. Live Titan enables Marketplace 1.7 at D:\UE_5.8\Engine\Plugins\Marketplace\PlanetGe5e8f23ed72d1V5\.<br>Pinned fork Plugins/PlanetGenPinned_1_4_0_RedMMO absent; RedPlanetGenCompat shims link without fork APIs | Fab/Epic Marketplace "PlanetGen" / CLM — install via Epic Launcher into UE 5.8                                                                         |
| PPG (Procedural Planet Generator)          | Clean RedMMO home world, ProfileV1, chunks/water/foilage | Off for fundamentals; Planet for clean RedMMO | Clean RedMMO.uproject will not generate home world. Fundamentals uproject has no PPG entry                                                                                                                                                     | Engine path seen: ... \Marketplace\Procedur890d9e860517V2\PPG.uplugin                                                                                  |
| FocalRig                                   | Rifle aim chain / AimWeapon ControlRig units             | Off — substituted                             | Plugin Enabled: false. Marketplace folder Intermediate-only (no FocalRig.uplugin). Stock ControlRig AimItem substitute preserves FocalAimTarget / FocalAimWeight                                                                               | Restore official Fab payload → enable in uproject → rebuild (Build.cs auto-links)                                                                      |
| WorldGen                                   | World-gen marketplace companion                          | Off                                           | Enabled: false; Intermediate-only; documented rollback                                                                                                                                                                                         | Same as FocalRig — restore WorldGen.uplugin then enable                                                                                                |
| Nwiro + NwirolntegrationKit                | Optional editor / environment kit                        | Opt                                           | Not needed for packaged play or fundamentals PIE                                                                                                                                                                                               | MarketplaceURL product 22e3bc90-ed91-4154-8bea-21c33e6a51b7 (Epic Fab deep link in Titan.uproject)                                                     |
| UnrealAllIntegrationPlatform (UAIP)        | Optional editor remote tools                             | Opt / blocked                                 | Enabled: false; payload Intermediate-only; skip Fab stall                                                                                                                                                                                      | Fab: <a href="https://www.fab.com/listings/0eedf909-00ac-4d95-b109-8fda51800fff">https://www.fab.com/listings/0eedf909-00ac-4d95-b109-8fda51800fff</a> |
| VibeMMOUiKit                               | VibeMMO player HUD kit (project Plugins/)                | Opt (HUD)                                     | HUD may fall back / miss widgets; gameplay C++ can still run                                                                                                                                                                                   | Project plugin (from Mac/Vibe handoff); keep with project                                                                                              |
| RedHUD                                     | Project HUD plugin                                       | Opt / project                                 | Status texture / fuel pixels incomplete without content                                                                                                                                                                                        | Project-owned under Plugins/RedHUD                                                                                                                     |
| SteamIntegrationKit                        | Steam OSS / sessions (SIK v1.9, App 480)                 | Opt (MP later)                                | Multiplayer lobby/Steam transport unavailable; PIE listen-server still possible with NULL OSS                                                                                                                                                  | Project Plugins/SteamIntegrationKit (~297 MB) — transfer deliberately                                                                                  |
| VibeUE                                     | Vibe tooling                                             | Off on Win64                                  | PlatformDenyList: Win64                                                                                                                                                                                                                        | Marketplace VibeUE581860d1833205V1                                                                                                                     |
| BpGeneratorUltimate                        | Blueprint generator tooling                              | Off on Win64                                  | Denied Win64                                                                                                                                                                                                                                   | Skip                                                                                                                                                   |
| OnlineSubsystemSteam / SteamSockets (Epic) | Legacy Steam                                             | Off                                           | Replaced by SIK                                                                                                                                                                                                                                | Keep disabled                                                                                                                                          |

## Content packs (Fab / Marketplace / purchased) — high level

| Pack / area                                                                              | Purpose                    | Fundamentals                                                  | If missing                                                                                                                                      |
|------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Action Trooper / Tall Female (Content/Action_Trooper, characters)                        | Playable trooper mesh/anim | Needed for full infantry feel; ThirdPerson pawn works without | Use ThirdPerson template character first                                                                                                        |
| StarSparrow                                                                              | Modular ship               | Planet/ship work                                              | Skip until planetary session                                                                                                                    |
| Projectiles Vol.1 / Muzzles / BeamsPack                                                  | Weapon VFX, grapple cable  | Nice-to-have for weapons polish                               | Bolt tracer C++ still compiles                                                                                                                  |
| Jet_Packs_Sci-Fi                                                                         | Jetpack meshes/VFX         | Jetpack presentation                                          | Logic may exist without mesh                                                                                                                    |
| SoStylized / Stylized Desert Oasis / Stylized Rocks / TropicalAlienWorld / Alien_* packs | Biome art                  | Planet art only                                               | Not needed for fundamentals map                                                                                                                 |
| Ultimate Stylized UI Crosshair                                                           | Arc sight                  | HUD polish                                                    | <a href="https://www.fab.com/listings/707a878d-0d11-431a-813c-a7aad10109f">https://www.fab.com/listings/707a878d-0d11-431a-813c-a7aad10109f</a> |
| SciFi_Skills_Icon                                                                        | Ability icons for Vibe HUD | HUD                                                           | See project cook notes if icons missing                                                                                                         |
| SpaceColony / AsteroidSpaceport / Cloudz_Hi5 / Vefects / etc.                            | Environment dressing       | No                                                            | Exclude from fundamentals                                                                                                                       |

Size note: Full Content/ on owner machine is tens of GB and is not all in git. Handoff repo ships ThirdPerson + git-whitelisted RedMMO crumbs + docs/source. Remaining packs travel via drive / separate sync.

## Quick “install only if you need...”

| You want...                | Install                                                       |
|----------------------------|---------------------------------------------------------------|
| Day-1 C++ / PIE / controls | Nothing from Fab — use TitanFundamentals.uproject             |
| Clean R92 planetary home   | PPG + clean project Content under WindowsData\Projects\RedMMO |
| Titan RedPlanetGen planet  | PlanetGen UE 5.8 Marketplace                                  |
| Official rifle aim rig     | FocalRig full Fab payload (not Intermediate leftovers)        |
| Steam friend sessions      | SteamIntegrationKit + Steam client + App 480                  |

## Owner-machine disposition snapshot (2026-08-07)

From R92\_DEPENDENCY\_DISPOSITION.md:

- FocalRig / WorldGen: not recoverable as official payloads → substituted / disabled
- PlanetGen pinned fork: absent; Marketplace 1.7 present; APIs shimmed
- TitanEditor Win64 Development: Succeeded after dependency unblock

# GitHub access — clone the handoff repo

## Clone (start here)

`https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git`

```
git clone https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff.git
cd RedMMOTitan-DeveloperHandoff
Read docs/DeveloperHandoff/START_HERE.md
start TitanFundamentals.uproject
```

Web UI: `https://github.com/addgamestudios-ops/RedMMOTitan-DeveloperHandoff`

This handoff repository is the same-repo collaboration surface for gameplay and environment: you own HubLogic / gameplay packages; artists own env map packages. See `MERGE_SAFE_WORLD.md`.

## What is included vs excluded

### Included

- Source/ (RedMMO gameplay C++)
- Config/
- docs/ DeveloperHandoff + EnvironmentArtistHandoff (MD + PDF)
- ProjectKnowledge/ (when present)
- Titan.uproject, TitanFundamentals.uproject, targets
- Content/ThirdPerson/ starter map
- Content/RedMMO/Maps/Hubs/ — Persistent / Env / Gameplay\_Logic stubs
- Selected Content/RedMMO/ crumbs when tracked

### Excluded

- Binaries/, Intermediate/, DerivedDataCache/, Saved/
- Full Fab content packs
- Plugins/SteamIntegrationKit (~297 MB) — optional; copy when doing Steam sessions
- Secrets beyond public Steam App ID 480 for Spacewar tests
- Owner-only workstation diagnostics and packaged build trees

## Private full Titan (optional later)

Historical / full private tree: `https://github.com/addgamestudios-ops/RedMMOTitan.git` — use only if you are granted access for content that is not in the public handoff clone. Day-1 HubLogic and netcode do not require it.